

Roll No.

ODD END SEMESTER EXAMINATION-2024

Name of the Course: B. Tech.

Name of the Paper: Engineering Physics

Time: 3 Hours

Semester: I

Paper Code: TPH 101

Maximum Marks: 100

Note:

- (i) All questions are compulsory
- (ii) Answer any two sub questions among a, b, & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each sub question carries 10 marks.

Q1	(20 marks)	CO 1
(a)	Discuss the phenomenon for formation of Newton's ring and derive the expression for diameters of bright rings.	
(b)	In Fraunhofer diffraction due to N-Slits derive the expression for resultant intensities.	
(c)	In Fresnel biprism experiment describe deviation method and displacement method to find the distance between two coherent virtual sources.	
Q2	(20 marks)	CO 2
(a)	Explain the population inversion and construction and working of three level laser with necessary energy level diagram.	
(b)	Discuss the production and detection of plane, elliptically, and circularly polarized light.	
(c)	Explain the working principle and construction of optical fiber and their applications.	
Q3	(20 marks)	CO 3
(a)	Deduce the expression for length contraction and time dilation using Lorentz transformation equations.	
(b)	State postulates for special theory of relativity and derive Einstein mass energy relation through relativity concept.	
(c)	A step index fiber has a numerical aperture of 0.26, a core of refractive index 1.5 and diameter of 100 μm , calculate: a) The refractive index of the cladding, b) the acceptance angle.	
Q4	(20 marks)	CO4 & CO5
(a)	Explain de-Broglie concept of matter waves. State Heisenberg uncertainty principle with relevant examples.	
(b)	Give the principle of nano-computing, its prospects and challenges.	
(c)	Derive Schrodinger time dependent wave equation also write the condition for Normalized wave function.	
Q5	(20 marks)	CO6, CO4 & CO5
(a)	Explain the phenomenon of superconductivity. Distinguish between Type I and Type II superconductors.	
(b)	Explain the concept of displacement current and derive Maxwell fourth equation.	
(c)	Give differences between quantum well, quantum wire, and quantum dots on the basis of energy level, quantum confinement and dimension.	

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